

Particle–hole conjugation is not Terwilliger’s $\dagger$: the symmetry that lifts is the one that buys nothing

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Abstract

Q151 asked whether particle–hole conjugation $\mathcal{C}$ of the fermionic Fock space is the antiautomorphism $\dagger$ of Terwilliger [4, §4], in the hope that [4, §4, `prop:threefourths`] would transfer to the classification of commuting ribbon operators and make half of the nonvanishing work free.

The answer is **no**, on two independent counts, and the reason is structural rather than technical. We prove (Theorem 2.1) that $\mathcal{C}$ conjugates the rank- e shape weight operator R_e^W to $R_e^{W^\vee}$, where $W^\vee$ is W precomposed with reversal-and-complementation of the occupancy word; the proof is three lines on the bead lattice and replaces a 1500-ribbon computational verification. But $X \mapsto \mathcal{C}X\mathcal{C}$ is conjugation by an involution, hence an *automorphism*, while Terwilliger’s $\dagger$ is by definition an *antiautomorphism*, and it is characterised by *fixing* the two generators of a Leonard pair, which $\mathcal{C}$ does not do. Separately, `prop:threefourths` turns out to say that any *three of four* tridiagonality conditions imply the fourth; it is not a statement about three quarters of anything, and the resemblance to `cor:quarter` of [3] is a collision of names.

What survives is sharper than what was asked for, and it is not what a first pass found. The admissibility system of [3, Rem. `rem:sym`] carries a group $(\mathbb{Z}/2)^2$ generated by reversal and complementation of words. We show (Theorems 4.4, 4.5) that exactly the subgroup generated by the composite $T = \text{cpl rev}$ is implemented by a Fock-space involution — no invertible graded linear map implements reversal alone (Proposition 4.7) — and that what this subgroup buys splits sharply in two. On the *nowhere-zero* part of the commuting locus it acts by a scalar $\kappa = (-1)^{e-1}\gamma(d/2)^{e/d}$ and buys nothing. On the *single-shape* part — the part that [1, Lem. 8.2] wrongly excluded and [2, 3] recovered — it acts as the free involution $w \mapsto w^T$ on ribbon shapes and halves the classification, its fixed points being exactly the antipalindromes, i.e. exactly the strong-condition family of [3, Thm. `thm:singleton2`]. So the Terwilliger *mechanism* does transfer; it is just not delivered by a $\dagger$, and it delivers only where the older classification was blind.

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1 Setting and statement

We use the normalisation of [1] throughout.

Convention 1.1 (Maya sets, moves, occupancy words). For a partition λ put $M(\lambda) = \{\lambda_j - j : j \geq 1\} \subset \mathbb{Z}$, a set containing all sufficiently negative integers and bounded above. A *legal rank- e move* in a Maya set M is a pair $b \mapsto b + e$ with $b \in M$ and $b + e \notin M$; it corresponds to adding a connected e -ribbon to λ . Its *occupancy word* is

$$u_M(b, e) = (u_1, \dots, u_{e-1}) \in \{0, 1\}^{e-1}, \quad u_i = [b + i \in M].$$

A *shape weight* of rank e over a commutative ring K is a function $W : \{0, 1\}^{e-1} \rightarrow K$; it defines

$$R_e^W |M\rangle = \sum_{\substack{b \in M \\ b+e \notin M}} W(u_M(b, e)) |(M \setminus \{b\}) \cup \{b+e\}\rangle,$$

equivalently $R_e^W s_\lambda = \sum_\mu W(u) s_\mu$ over e -ribbons μ/λ .

Convention 1.2 (the three word involutions). On $\{0, 1\}^n$ define

$$(\text{rev } u)_i = u_{n+1-i}, \quad (\text{cpl } u)_i = 1 - u_i, \quad u^T := \text{cpl rev } u = \text{rev cpl } u, \quad (u^T)_i = 1 - u_{n+1-i}.$$

For a shape weight W of rank e we write $W^\vee(u) := W(u^T)$, the weight precomposed with reversal-and-complementation on $\{0, 1\}^{e-1}$. By [1, Lem. `lem:transpose`], $u \mapsto u^T$ is transposition of the ribbon.

Convention 1.3 (particle-hole conjugation). For a Maya set M put

$$\mathcal{C}(M) := \{-1 - x : x \in \mathbb{Z} \setminus M\}.$$

This is again a Maya set of charge 0, and $\mathcal{C}^2 = \text{id}$. We also write $\mathcal{C}$ for the induced linear involution $|M\rangle \mapsto |\mathcal{C}M\rangle$ of the charge-0 Fock space.

Lemma 1.4. $\mathcal{C}(M(\lambda)) = M(\lambda')$: *particle-hole conjugation is transposition of partitions.*

Proof. Classical. Reading $\epsilon_x = [x \in M(\lambda)]$ left to right along $\mathbb{Z}$ gives the boundary lattice path of λ , with $\epsilon_x = 1$ a vertical step and $\epsilon_x = 0$ a horizontal step, normalised so that charge 0 puts the corner of the staircase between -1 and 0 . Transposing λ reflects the diagram in the main diagonal, which reverses the path about that point and exchanges the two step types; on ϵ this is exactly $\epsilon_x \mapsto 1 - \epsilon_{-1-x}$, i.e. $M \mapsto \mathcal{C}(M)$. Verified independently on all 97 partitions of size ≤ 9 (Check V1 of §5). $\square$

2 Theorem A: what $\mathcal{C}$ does to a ribbon operator

Theorem 2.1. *Let $e \geq 1$ and let W be a shape weight of rank e over any commutative ring. Then*

$$\mathcal{C} R_e^W \mathcal{C} = R_e^{W^\vee}, \quad W^\vee(u) = W(u^T).$$

Proof. Fix a Maya set M and write $M^c = \mathcal{C}(M)$, so that for every $x \in \mathbb{Z}$

$$x \in M \iff -1 - x \notin M^c. \quad (1)$$

Moves correspond. Let $b \mapsto b + e$ be a legal move in M and put $b^c := -1 - b - e$. By (1), $b \in M$ gives $-1 - b \notin M^c$, i.e. $b^c + e \notin M^c$; and $b + e \notin M$ gives $-1 - b - e \in M^c$, i.e. $b^c \in M^c$. So $b^c \mapsto b^c + e$ is a legal rank- e move in M^c . The assignment $b \mapsto -1 - b - e$ is an injection, and applying the same argument to M^c (using $\mathcal{C}^2 = \text{id}$) produces the inverse; so it is a bijection between the legal rank- e moves of M and those of M^c .

Words reverse-and-complement. For $1 \leq i \leq e - 1$,

$$u_{M^c}(b^c, e)_i = [b^c + i \in M^c] = [-1 - (b + e - i) \in M^c] \stackrel{(1)}{=} [b + e - i \notin M] = 1 - u_M(b, e)_{e-i},$$

where $e - i$ runs over $1, \dots, e - 1$ as i does. That is exactly $u_{M^c}(b^c, e) = u_M(b, e)^T$.

Targets correspond. Write $M' = (M \setminus \{b\}) \cup \{b + e\}$. The set M' differs from M only at b and $b + e$, with both memberships flipped; by (1), $\mathcal{C}M'$ differs from M^c only at $-1 - b = b^c + e$ and $-1 - b - e = b^c$, again with both flipped. Hence $\mathcal{C}M' = (M^c \setminus \{b^c\}) \cup \{b^c + e\}$, the target of the corresponding move.

Now compute. Since $\mathcal{C}^2 = \text{id}$,

$$\mathcal{C}R_e^W \mathcal{C} |M\rangle = \mathcal{C}R_e^W |M^c\rangle = \mathcal{C} \sum_{b^c} W(u_{M^c}(b^c, e)) |\cdots\rangle = \sum_b W(u_M(b, e)^T) |M'\rangle = R_e^{W^\vee} |M\rangle. \quad \square$$

Remark 2.2 (a computational verification becomes a proof). [1, Lem. `lem:transpose`] — transposition acts on occupancy words by reversal-and-complementation — was recorded there as *verified* against 1500 ribbons, with no derivation. The middle paragraph of the proof above is a derivation, and it never mentions Young diagrams: (1) does all the work. Combined with Lemma 1.4 it upgrades `lem:transpose` from `computed` to `proved`.

Corollary 2.3. $X \mapsto \mathcal{C}X\mathcal{C}$ is an algebra automorphism of End of the Fock space, and it carries $R_e^W \mapsto R_e^{W^\vee}$ for every e and W . In particular, for weights of ranks e, f ,

$$\mathcal{C} (R_e^W R_f^{\bar{W}}) \mathcal{C} = R_e^{W^\vee} R_f^{\bar{W}^\vee},$$

with the factors in the same order.

Proof. Conjugation by an involution is multiplicative: $\mathcal{C}XY\mathcal{C} = (\mathcal{C}X\mathcal{C})(\mathcal{C}Y\mathcal{C})$. Apply Theorem 2.1 to each factor. $\square$

3 Q151, answered

3.1 Terwilliger's $\dagger$, from its definition

We quote [4, §4] verbatim in substance. Let $\Phi = (A; \{E_i\}; A^*; \{E_i^*\})$ be a pre Leonard system on V with $\dim V = d + 1 < \infty$, and assume $E_i^* A E_j^* = 0$ for $|i - j| > 1$ and $\neq 0$ for $|i - j| = 1$. An *antiautomorphism* of $\text{End}(V)$ is an $\mathbb{F}$ -linear bijection ζ with $(XY)^\zeta = Y^\zeta X^\zeta$ for all X, Y . Then [4, Lem. `lem:antiaut`]:

there exists a *unique antiautomorphism* $\dagger$ of $\text{End}(V)$ that *fixes each* A, A^* ; it fixes everything in $\mathcal{D}$ and $\mathcal{D}^*$, fixes each E_i, E_i^* , and $(X^\dagger)^\dagger = X$.

Two features of this definition are load-bearing, and $\mathcal{C}$ has neither.

Proposition 3.1 (variance). $X \mapsto CXC$ is multiplicative, not antimultiplicative, on the algebra generated by the ribbon operators. Concretely, for generic weights $W, \bar{W}$ of ranks e, f ,

$$\mathcal{C}(R_e^W R_f^{\bar{W}}) \mathcal{C} = R_e^{W^\vee} R_f^{\bar{W}^\vee} \neq R_f^{\bar{W}^\vee} R_e^{W^\vee},$$

the two candidates differing by $[R_e^{W^\vee}, R_f^{\bar{W}^\vee}]$.

Proof. The equality is Corollary 2.3. For the inequality it suffices to exhibit one pair of ranks and one λ with $[R_e^{W^\vee}, R_f^{\bar{W}^\vee}]_{s_\lambda} \neq 0$ for generic $W, \bar{W}$; by [1, Thm. `thm:main`] the commuting locus is a proper closed subvariety of weight space whenever it is nonempty, so generic weights do not commute. Check V3 of §5 exhibits the witnesses explicitly. $\square$

Remark 3.2 (naming the separator in advance). An automorphism and an antiautomorphism agree on every product of length ≤ 1 . No computation with a *single* ribbon operator can tell them apart, and — the point that had to be written down before running anything — *on the commuting locus the two candidates coincide*, because $[A, B] = 0$ makes $AB = BA$. A test performed with Murnaghan–Nakayama weights, or with any pair drawn from the locus of [1, Thm. `thm:main`], is constant in the direction it is supposed to measure. The separating coordinate is a product of two operators with *independent generic* weights. Check V3 runs there and reports how many λ actually separate, so that the control is not silently empty.

Proposition 3.3 (the fixing property). $\dagger$ is characterised among antiautomorphisms by fixing each of A, A^* . The analogous statement fails for $\mathcal{C}$: by Theorem 2.1, $\mathcal{C}R_e^W \mathcal{C} = R_e^W$ if and only if $W^\vee = W$, i.e. W is invariant under ribbon transposition, and for $e \geq 2$ that is a proper subspace of weight space, of dimension 2^{e-2} for e even and $2^{e-2} + 2^{(e-3)/2}$ for e odd.

Proof. Put $n = e - 1$. The fixed space of $W \mapsto W \circ T$ on functions $\{0, 1\}^n \rightarrow K$ has dimension equal to the number of orbits of T on $\{0, 1\}^n$, namely $\frac{1}{2}(2^n + \#\text{Fix}(T))$. A word is fixed iff $u_i + u_{n+1-i} = 1$ for all i . If n is odd the index $i = (n + 1)/2$ gives $2u_i = 1$, which is impossible, so $\#\text{Fix}(T) = 0$; if n is even, u is free on $i \leq n/2$ and forced on the rest, so $\#\text{Fix}(T) = 2^{n/2}$. Hence

$$\dim\{W : W^\vee = W\} = \begin{cases} 2^{e-2}, & e \text{ even,} \\ 2^{e-2} + 2^{(e-3)/2}, & e \text{ odd,} \end{cases}$$

which is $< 2^{e-1}$ for every $e \geq 2$. $\square$

3.2 `prop:threefourths` is a collision of names

Remark 3.4. [4, `prop:threefourths`] reads, in full: given four conditions (i)–(iv) asserting one-sided tridiagonality of A with respect to $\{E_i^*\}$ and of A^* with respect to $\{E_i\}$, *if at least three of (i)–(iv) hold then all four hold*, i.e. the pre Leonard system is a Leonard system. “Three fourths” counts *conditions*, not *shapes*. `cor:quarter` of [3] says that a nonzero commuting weight is nowhere zero or vanishes on more than three quarters of the rank- d shapes. The two statements share a fraction and nothing else. There is no theorem to transfer.

Remark 3.5 (what the mechanism actually is, and why it still does not transfer). The *proof* of `prop:threefourths` is one line and is the thing worth naming: $\dagger$ is an antiautomorphism fixing A, A^* , hence $(E_i A^* E_j)^\dagger = E_j A^* E_i$, hence condition (iii) holds if and only if (iv) does. The pattern is: *an involution that pairs up the nonvanishing conditions makes half of them free*. That is exactly

the hope Q151 was built on, and Theorem 4.4 below is the reason it fails here: the involution available to us acts on the commuting locus by a scalar, so the pairing it induces on nonvanishing conditions is the trivial pairing, and half of nothing is nothing.

Theorem 3.6 (Q151). *Particle-hole conjugation $\mathcal{C}$ is not Terwilliger's $\dagger$. It fails antimultiplicativity (Proposition 3.1) and it fails the fixing property that characterises $\dagger$ (Proposition 3.3); and the proposition whose transfer motivated the question is about a different subject (Remark 3.4).*

4 Theorem C: which symmetry lifts, and what it is worth

Remark 4.1 (the locus I must work over). [1, Thm. `thm:main`] describes the commuting locus as the multiplicative gcd-periodic characters W^γ . That description is *superseded*: [2, Cor. `cor:main0`] shows that [1, Lem. 8.2] — nonvanishing of a commuting weight — is false for $\gcd(e, f) \geq 3$, and [2, Thm. `thm:main0`] replaces the classification by: for $W \neq 0 \neq \bar{W}$ and $d = \gcd(e, f)$,

$$[R_e^W, R_f^{\bar{W}}] = 0 \iff \sigma = c\tau^{\otimes(e/d)}, \quad \bar{\sigma} = \bar{c}\tau^{\otimes(f/d)}, \quad \sigma \text{ satisfies } (2B_e^*), \quad (2)$$

for some nonzero rank- d shape weight τ over F , where $\sigma(u) = (-1)^{|u|}W(u)$ and $\otimes$ is concatenation of shape weights [2, Conv. `conv:otimes`]. The W^γ are exactly the nowhere-zero points of (2); the points with zeros are the new ones, and they are where the answer to Q151 stops being trivial. A first pass at this section, written against [1, Thm. `thm:main`], concluded the opposite of Theorem 4.5(b) below.

[3, Rem. `rem:sym`] records that admissibility of a support system is preserved by `cpl` and by `rev`, and that the composite is ribbon transposition T ; the remark after [3, Cor. `cor:48`] records that the list of admissible single shapes is stable under T . Both are statements about *words*. Theorem 2.1 promotes one of them to a statement about *operators*, and that is what makes the accounting below possible.

Lemma 4.2 (the involutions pass through $\otimes$). *Let τ be a shape weight of rank d , $k \geq 1$, and $g \in \{\text{rev}, \text{cpl}, T\}$ acting on words of the appropriate length. Then $(\tau^{\otimes k}) \circ g = (\tau \circ g)^{\otimes k}$.*

Proof. Write $n = kd - 1$. In $(\tau^{\otimes k})(y) = \prod_{i=1}^k \tau(y_{[(i-1)d+1, id-1]})$ the $k-1$ letters in positions $d, 2d, \dots, (k-1)d$ are ignored. Under $i \mapsto n+1-i = kd-i$, position $(i-1)d+j$ with $1 \leq j \leq d-1$ goes to $(k-i)d+(d-j)$, i.e. block i goes to block $k+1-i$ with its letters reversed, and the ignored positions id go to the ignored positions $(k-i)d$. So `rev` permutes the blocks and reverses each; since the product over blocks is a product of scalars it is insensitive to the order of the blocks, giving $(\tau^{\otimes k}) \circ \text{rev} = (\tau \circ \text{rev})^{\otimes k}$. For `cpl` the block positions are fixed and each block is complemented. Compose for T . Verified symbolically for all $d \leq 4$, $k \leq 3$ with $kd-1 \leq 9$ (1368 identities, 0 failures; Check V7). $\square$

Lemma 4.3 (the two-bead condition is G -stable). *Let σ be a shape weight of rank m satisfying $(2B_m^*)$. Then $\sigma \circ \text{rev}$ and $\sigma \circ \text{cpl}$ satisfy it, hence so does $\sigma \circ T$.*

Proof. Write the condition as: for all $\delta \in [1, m-1]$ and all $\pi, \varsigma \in \{0, 1\}^{\delta-1}$, $\pi' \in \{0, 1\}^{m-\delta-1}$, $\sigma(\pi 0 \pi') \sigma(\pi' 0 \varsigma) = \sigma(\pi 1 \pi') \sigma(\pi' 1 \varsigma)$.

Complementation. $(\sigma \circ \text{cpl})(\pi b \pi') = \sigma(\bar{\pi} \bar{b} \bar{\pi}')$, so the b -side for $\sigma \circ \text{cpl}$ at $(\delta; \pi, \pi', \varsigma)$ equals the $\bar{b}$ -side for σ at $(\delta; \bar{\pi}, \bar{\pi}', \bar{\varsigma})$. The hypothesis equates the two sides for σ , hence for $\sigma \circ \text{cpl}$.

Reversal. $(\sigma \circ \text{rev})(\pi b \pi') = \sigma(\tilde{\pi}' b \tilde{\pi})$ where $\tilde{}$ is reversal. Hence

$$(\sigma \circ \text{rev})(\pi b \pi') (\sigma \circ \text{rev})(\pi' b \varsigma) = \sigma(\tilde{\pi}' b \tilde{\pi}) \sigma(\tilde{\varsigma} b \tilde{\pi}') = \sigma(\tilde{\varsigma} b \tilde{\pi}') \sigma(\tilde{\pi}' b \tilde{\pi}),$$

the last step being commutativity of the ring. The right-hand side is the b -side of the condition for σ at the *same* δ , with $(\pi, \pi', \varsigma) := (\tilde{\varsigma}, \tilde{\pi}', \tilde{\pi})$ — the lengths match, since $|\tilde{\varsigma}| = |\tilde{\pi}| = \delta - 1$ and $|\tilde{\pi}'| = m - \delta - 1$. Apply the hypothesis. Verified exhaustively over $\mathbb{F}_5$ for $m = 3, 4$ (all 25 resp. 81 solutions; 0 failures; Check V8). $\square$

Theorem 4.4 (the symmetry group acts, and T is the part an operator sees). *Let $1 \leq e < f$, $d = \gcd(e, f)$. The group $G = \langle \text{rev}, \text{cpl} \rangle \cong (\mathbb{Z}/2)^2$ acts on the commuting locus (2) by $\tau \mapsto \tau \circ g$ on the rank- d seed. Under this action the subgroup $\langle T \rangle$ is implemented on operators: by Theorem 2.1,*

$$\mathcal{C} R_e^W \mathcal{C} = R_e^{W^\vee}, \quad \mathcal{C} R_f^{\bar{W}} \mathcal{C} = R_f^{\bar{W}^\vee},$$

and $(W, \bar{W})$ commute if and only if $(W^\vee, \bar{W}^\vee)$ do.

Proof. Lemma 4.2 shows g carries $\tau^{\otimes k}$ to $(\tau \circ g)^{\otimes k}$, so the shape $\sigma = c\tau^{\otimes(e/d)}$, $\bar{\sigma} = \bar{c}\tau^{\otimes(f/d)}$ of (2) is preserved with seed $\tau \circ g$, up to the global sign $(-1)^{m-1}$ incurred by cpl in passing between W and σ (which is a unit and does not affect (2)). Lemma 4.3 preserves $(2B_e^*)$. The operator statement is Corollary 2.3 applied to both factors. $\square$

Theorem 4.5 (the two regimes). (a) The nowhere-zero part. *Let τ be nowhere zero, so that $W = \alpha W_e^\gamma$, $\bar{W} = \beta W_f^\gamma$ in the notation of [1, Conv. conv:gamma]. Then*

$$W_e^\gamma \circ \text{rev} = W_e^{\gamma^{-1}}, \quad W_e^\gamma \circ T = \kappa W_e^\gamma, \quad \kappa = (-1)^{e-1} \Gamma, \quad \Gamma = \begin{cases} 1, & d \text{ odd}, \\ \gamma(d/2)^{e/d} \in \{\pm 1\}, & d \text{ even}. \end{cases}$$

So T acts by a scalar — trivially, modulo the scalars α, β that (2) quotients by — while rev acts by $\gamma \mapsto \gamma^{-1}$, which moves points if and only if $d \geq 3$.

(b) The single-shape part. *Let $\tau = c\mathbb{K}_w$ with $w \in \{0, 1\}^{d-1}$ satisfying (NB) of [3, Def. def:border] — that is, writing $n = d - 1$ and calling $p \in [0, n - 1]$ a border length when $w_{[1,p]} = w_{[n-p+1,n]}$ (so $p = 0$ always is one), $w_{p+1} \neq w_{n-p}$ for every border length p . Then $\tau \circ T = c\mathbb{K}_{w^T}$, and w^T again satisfies (NB). Here T does not act by a scalar: the smallest instance is $d = 4$, where $w = 110 \mapsto w^T = 100$, i.e. the ribbon shape $(2, 1, 1)$ is exchanged with $(3, 1)$. The fixed points of T are exactly the antipalindromes, hence exactly the w for which $\mathbb{K}_w$ satisfies the strong condition $(2B_d^{**})$ [3, Thm. thm:singleton2].*

Proof. (a) Write $\sigma_e^\gamma(u) = \prod_j \gamma(j \bmod d)^{u_j}$, so $W_e^\gamma = (-1)^{|u|} \sigma_e^\gamma$. Then $\sigma_e^\gamma(\text{rev } u) = \prod_j \gamma(j)^{u_{e-j}} = \prod_i \gamma(e - i)^{u_i}$, and $d \mid e$ gives $\gamma(e - i \bmod d) = \gamma(-i \bmod d) = \gamma(i)^{-1}$, so this is $\sigma_e^{\gamma^{-1}}(u)$; as $|\text{rev } u| = |u|$, the first identity follows. Also $\sigma_e^\gamma(\text{cpl } u) = \prod_j \gamma(j)^{1-u_j} = \Gamma \sigma_e^{\gamma^{-1}}(u)$ and $|\text{cpl } u| = e - 1 - |u|$, so $W_e^\gamma \circ \text{cpl} = (-1)^{e-1} \Gamma W_e^{\gamma^{-1}}$; composing gives $W_e^\gamma \circ T = (-1)^{e-1} \Gamma W_e^\gamma$. For Γ : grouping $j = 1, \dots, e - 1$ into e/d full residue systems and using $\gamma(0) = 1$, $\Gamma = \left(\prod_{\delta \in \mathbb{Z}/d} \gamma(\delta) \right)^{e/d}$; the relation $\gamma(\delta)\gamma(-\delta) = 1$ kills every two-element orbit, leaving $\gamma(0) = 1$ and, for d even, $\gamma(d/2)$, which squares to 1. Finally $\gamma \mapsto \gamma^{-1}$ is the identity on the whole parameter group iff that group is 2-torsion; by [1, Cor. cor:size] it is $(F^\times)^{\lfloor (d-1)/2 \rfloor} \times \mu_2(F)^{\lfloor 2|d \rfloor}$, which is 2-torsion iff $\lfloor (d-1)/2 \rfloor = 0$, i.e. $d \leq 2$.

(b) $(\mathbb{K}_w \circ T)(u) = [u^T = w] = [u = w^T]$ since $T^2 = \text{id}$. That w^T satisfies (NB) is Lemma 4.3 together with [3, Thm. thm:singleton]; directly, T preserves border lengths ($w_{[1,p]}^T$ is the complemented reverse of $w_{[n-p+1,n]}$ and vice versa) and $w_{p+1}^T = 1 - w_{n-p}$, $w_{n-p}^T = 1 - w_{p+1}$, so the clause $w_{p+1} \neq w_{n-p}$ is carried to itself. The witness $d = 4$ is Check V9. For the fixed points: $w = w^T$ says $w_i = 1 - w_{n+1-i}$ for all i , i.e. $w_i \neq w_{n+1-i}$, which is the antipalindrome condition of [3, Thm. thm:singleton2]. $\square$

Corollary 4.6 (the parity, explained). *T has no fixed point on $\{0, 1\}^n$ for n odd, and exactly $2^{n/2}$ for n even. Hence the (2B**) single-shape family is empty for d even and has $2^{(d-1)/2}$ members for d odd — recovering the count of [3, Thm. `thm:singleton2`] from the fixed-point count of particle-hole conjugation alone.*

Proof. $w = w^T$ reads $w_i + w_{n+1-i} = 1$. For n odd the index $i = (n+1)/2$ demands $2w_i = 1$, impossible. For n even, w is free on $i \leq n/2$ and determined on the rest. Now $n = d - 1$. $\square$

Proposition 4.7 (only the composite lifts). *There is no invertible degree-preserving linear map $\mathcal{J}$ of the Fock space with*

$$\mathcal{J} R_e^W \mathcal{J}^{-1} = R_e^{W^{\text{orev}}} \quad \text{for all } e \geq 1 \text{ and all shape weights } W.$$

The same holds with rev replaced by cpl.

Proof. Suppose such a $\mathcal{J}$ exists. Degree 0 of the Fock space is spanned by $s_\emptyset$, so $\mathcal{J} s_\emptyset = c s_\emptyset$ with $c \neq 0$.

Take $e = 3$ and let δ_u denote the weight supported on the single word u with value 1. Under [1, Lem. `lem:dict`] the word $(1, 0)$ is the ribbon with row lengths $(2, 1)$ read top first, and $(0, 1)$ is the ribbon $(1, 2)$; and $\text{rev}(1, 0) = (0, 1)$. Hence the hypothesis gives $\mathcal{J} R_3^{\delta_{(1,0)}} = R_3^{\delta_{(0,1)}} \mathcal{J}$. Apply both sides to $s_\emptyset$. A ribbon added to $\emptyset$ is a hook, so its rows are weakly decreasing top to bottom; the shape $(1, 2)$ is not, and the shape $(2, 1)$ is. Therefore $R_3^{\delta_{(0,1)}} s_\emptyset = 0$ while $R_3^{\delta_{(1,0)}} s_\emptyset = s_{(2,1)}$, so $\mathcal{J} s_{(2,1)} = R_3^{\delta_{(0,1)}}(c s_\emptyset) = 0$, contradicting injectivity of $\mathcal{J}$. For cpl the same two words work, since $\text{cpl}(1, 0) = (0, 1)$ as well. $\square$

Remark 4.8 (the shape of the answer). $R_3^{\delta_{(0,1)}}$ is not the zero operator — it sends $s_{(1)} \mapsto s_{(2,2)}$ — so the obstruction is not that one of the two tromino operators vanishes. It is that they vanish on *different* vectors, and the vacuum separates them. This is the smallest live case: for $e \leq 2$ the involution rev is the identity on $\{0, 1\}^{e-1}$, so nothing can be seen there.

Corollary 4.9 (the honest accounting). *Of the $(\mathbb{Z}/2)^2$ available on words, exactly the subgroup $\langle T \rangle$ is implemented by a Fock-space involution (Theorem 2.1, Proposition 4.7); and what that subgroup buys depends on where one stands on the locus (2):*

- on the nowhere-zero part it acts by the scalar κ , so $\text{supp}(W)$ is already T -stable and no nonvanishing condition is paired with a different one: nothing is free;
- on the single-shape part — the part that is new, and that [1, Lem. 8.2] wrongly excluded — it acts as the involution $w \mapsto w^T$, whose fixed points are the $2^{(d-1)/2}$ antipalindromes when d is odd and none when d is even. So the classification of single-shape commuting pairs need only be carried out on T -orbit representatives, and half of it is free: the orbit counts are 1, 2, 2, 6, 6, 18, 24, 60 for $d = 2, \dots, 9$ against 0, 2, 4, 8, 12, 28, 48, 104 shapes.

In particular the mechanism of [4, `prop:threefourths`] (Remark 3.5) does have an analogue here — but it is delivered by an automorphism, not an antiautomorphism, and it delivers on exactly the half of the locus that the superseded [1, Thm. `thm:main`] did not see.

Remark 4.10 (a terminology hazard, recorded). [3, Def. `def:border`] defines “non-overlapping” to mean (NB): every border length p of w has $w_{p+1} \neq w_{n-p}$. This is weaker than the standard meaning of “non-overlapping” (= unbordered = no proper nonempty border) in combinatorics on words, and the two differ from $d = 5$ on: at $d = 5$ there are 8 words satisfying (NB) and only 6

unbordered ones, the difference being 0101 and 1010. The counts 0, 2, 4, 8, 12, 28, 48, 104 of [3, §ver] are the (NB) counts. Any write-up that merges these sources must not import the word; it must import the condition.

5 Verification

All checks are in `scratch/q151/` (`maya.py`, `sep.py`), exact arithmetic over $\mathbb{Q}$ or $\mathbb{Q}(\gamma)$ via `SYMPY`; no floating point.

V1 ($\mathcal{C}$ is transposition). For all 97 partitions of size ≤ 9 , $\mathcal{C}(M(\lambda)) = M(\lambda')$. 0 failures.

V2 (Theorem 2.1 at move level). For all 97 partitions of size ≤ 9 and all $e \leq 6$: the map $b \mapsto -1 - b - e$ is a bijection from the legal rank- e moves of $M(\lambda)$ to those of $\mathcal{C}M(\lambda)$; the occupancy word transforms by $u \mapsto u^T$; and the targets correspond under $\mathcal{C}$. 2670 moves, 0 failures.

V3 (the separator; Prop. 3.1). With $W, \bar{W}$ independent generic symbolic weights — $2^{e-1} + 2^{f-1}$ free symbols, deliberately off the commuting locus — and for every partition λ with $|\lambda| \leq 5$:

(e, f)	$\mathcal{C}(R_e R_f) \mathcal{C} = R_e^\vee R_f^\vee$	$\mathcal{C}(R_e R_f) \mathcal{C} = R_f^\vee R_e^\vee$	$\#\{\lambda : \text{the two candidates differ}\}$
(1, 2)	19/19	0/19	19/19
(2, 3)	19/19	0/19	19/19
(2, 2)	19/19	0/19	15/19
(3, 4)	19/19	0/19	19/19
(2, 4)	19/19	0/19	19/19

The last column is the positive control demanded by Remark 3.2: the test is not constant in the direction it measures.

V4 (κ ; Thm. 4.4(ii)). For every $d \leq 7$, every $e \leq 12$ with $d \mid e$, and each choice $\gamma(d/2) = \pm 1$ when d is even, with γ otherwise a free symbol on orbit representatives: $W_e^\gamma(u^T) = \kappa W_e^\gamma(u)$ for all u , with $\kappa = (-1)^{e-1} \gamma(d/2)^{e/d}$. 41 triples $(d, e, \pm)$, 0 failures.

V5 (an untuned witness at $d = 3$). $(e, f) = (3, 6)$, $\gamma = (1, 3, \frac{1}{3})$ on $\mathbb{Z}/3$ — a point of the commuting locus that is *not* Murnaghan–Nakayama, and for which $\gamma \neq \gamma^{-1}$. Then $[R_3^W, R_6^{\bar{W}}]_{s_\lambda} = 0$ for all 12 partitions λ with $|\lambda| \leq 4$; likewise for γ^{-1} . Directly: $W \circ \text{rev} = W^{\gamma^{-1}}$ and $\bar{W} \circ \text{rev} = \bar{W}^{\gamma^{-1}}$, while $W \circ T = +W$ and $\bar{W} \circ T = -\bar{W}$, matching $\kappa = (-1)^2 = 1$, $\bar{\kappa} = (-1)^5 = -1$. This is the separator of Theorem 4.4(iii): reversal moves the point, the composite does not.

V6 (Prop. 4.7). $R_3^{\delta_{(0,1)}} s_\emptyset = 0$, $R_3^{\delta_{(1,0)}} s_\emptyset = s_{(2,1)}$, $R_3^{\delta_{(0,1)}} s_{(1)} = s_{(2,2)}$.

V7 (Lemma 4.2). With τ a generic symbolic rank- d weight: for all $d \leq 4$, $k \leq 3$ with $kd - 1 \leq 9$, and $g \in \{\text{rev}, \text{cpl}, T\}$, $(\tau^{\otimes k})(gy) = (\tau \circ g)^{\otimes k}(y)$ for every word y . 1368 identities, 0 failures.

V8 (Lemma 4.3). Exhaustive over $\mathbb{F}_5$: all 25 solutions of $(2B_3^*)$ and all 81 of $(2B_4^*)$ remain solutions after rev , cpl and T . 0 failures.

V9 (Thm. 4.5(b); the separator for “ T acts by a scalar”). Exhaustive over $\mathbb{F}_5$ and $\mathbb{F}_7$, $m = 3, 4$: every nonzero $(2B_3^*)$ weight satisfies $\sigma \circ T \in F^\times \sigma$, but at $m = 4$ exactly 32 of 80 (over

$\mathbb{F}_5$) and 72 of 168 (over $\mathbb{F}_7$) do not — the failures being the single-shape weights on non- T -symmetric words, e.g. $\sigma = \mathbb{K}_{110}$ with $\sigma \circ T = \mathbb{K}_{100}$. **This is the check that refuted the first version of this section**, which had asserted the scalar action on the whole locus on the strength of the superseded [1, Thm. `thm:main`].

V10 (the (NB) family and its T -orbits). Direct enumeration for $d \leq 9$: $\{w : \mathbb{K}_w \text{ satisfies } (2B_d^*)\}$ agrees with (NB) of [3, Def. `def:border`] in all $\sum_d 2^{d-1}$ cases, with counts 0, 2, 4, 8, 12, 28, 48, 104 for $d = 2, \dots, 9$ — matching [3, §ver]. The family is rev-, cpl- and T -stable; $\text{Fix}(T)$ inside it equals the antipalindromes exactly, with counts 0, 2, 0, 4, 0, 8, 0, 16; the T -orbit counts are 1, 2, 2, 6, 6, 18, 24, 60. Separately, the (NB) family strictly contains the *unbordered* words from $d = 5$ on (8 versus 6), which is Remark 4.10.

6 What is not proved

1. Proposition 4.7 rules out invertible *degree-preserving* linear implementations of rev. It does not rule out an implementation by a map that mixes degrees, nor one defined only on a subquotient. I believe no such map exists but have not proved it; the hypothesis is used exactly once, to force $\mathcal{J}s_\emptyset \in \mathbb{F}^\times s_\emptyset$.
2. Theorem 4.5 treats two parts of the locus (2): the nowhere-zero seeds and the single-shape seeds. These do not exhaust it — a seed τ may have zeros without being supported on one word — and for the intermediate seeds I have not determined when $\tau \circ T \in F^\times \tau$. What Check V9 does establish is that the scalar action fails somewhere, which is all Corollary 4.9 needs; the precise T -fixed locus of (2) is open.
3. Corollary 4.9’s “half of it is free” is an accounting statement about a classification, not a proof that any particular proof shortens. Concretely: for each T -orbit $\{w, w^T\}$ of size 2, Theorem 2.1 gives the second commuting pair from the first by conjugation; it does not give the *verification* that w satisfies (NB) for free, since that was never the expensive half.
4. The Leonard-system analogy is now closed in the negative for $\dagger$, but not for the rest of [4]: the finding of `sources.json` that PA2 (nonvanishing) and PA5 (rigidity) appear there as *independent* conditions remains the live warning, and is untouched by anything above.

References

- [1] Clio, *Shape weights on ribbons: the commuting locus is the transpose-equivariant $\gcd(e, f)$ -periodic characters*, `proofs/2026-09-15-c1-Q149-shape-weights.tex`, 15 September 2026.
- [2] Clio, *Q150: the nonvanishing lemma is false*, `proofs/2026-09-16-c1-Q150-nonvanishing.tex`, 16 September 2026.
- [3] Clio, *Q150 (H2): single-shape commuting operators*, `proofs/2026-09-16-c2-Q150-H2-single-shape.tex`, 16 September 2026.
- [4] P. Terwilliger, *Notes on the Leonard system classification*, `arXiv:2003.09668`; Graphs and Combinatorics **37** (2021) 1687–1748. Definitions and statements above are quoted from the arXiv LaTeX source (`arxiv.org/e-print/2003.09668`), §4 (`lem:antiaut`) and `prop:threefourths`.